J. Dairy Sci. 106:4545–4558
https://doi.org/10.3168/jds.2022-22476
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Growth performance, blood metabolites, ruminal fermentation, and bacterial community in preweaning dairy calves fed corn silage–included starter and total mixed ration

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ABSTRACT

The objective of this study was to evaluate the effects of the inclusion of whole-plant corn silage (WPCS) in a starter or total mixed ration (TMR) on growth, blood metabolites, ruminal fermentation, and microbial community in preweaning dairy calves. A total of 45 healthy dairy calves were blocked by date of birth and randomly assigned to 1 of 3 treatments: 100% calf starter (CONS), a mix of 85% calf starter and 15% WPCS [dry matter (DM) basis; CSCS], or 100% WPCS-based lactation TMR (CTMR). Pasteurized normal milk was fed to all the animals under the same regimen. The experiment ran from when the calves were 2 d old to weaning at 63 d. Milk and feed intakes were recorded daily. Growth performance data and blood samples were collected on wk 3, 5, 7, and 9 of the experiment. Rumen fluid was sampled at 40 and 60 d. The 3 treatments had different particle size fractions. The CSCS group had greater medium fraction (<19 mm, >8 mm) and particles retained on 8-mm sieves than the other 2 groups, whereas the CTMR group had the greatest long (>19 mm) and fine (<4 mm) fractions and physically effective neutral detergent fiber (NDF) on 8- and 4-mm sieves, but had the smallest short fraction (<8 mm, >4 mm) and particles retained on 4-mm sieves. The 24-h in vitro digestibility of DM, crude protein (CP), NDF, and acid detergent fiber (ADF) were decreased in order by the CONS, CSCS, and CTMR groups. Compared with the CONS group, the digestibility of ether extract (EE) was lower in the CSCS and CTMR groups, whereas the digestibility of starch was similar among treatments. During the experimental period, the DM, CP, and metabolizable energy intakes from milk, solid feed, and total feed were not affected by treatments. The NDF, ADF, and EE intakes and potentially digestible intakes were greater in the CTMR group than in the other 2 groups. With the exception that body barrel was greater for calves fed CSCS, growth parameters and blood metabolites were similar among treatments. Compared with the CSCS group, the CTMR group had greater rumen pH and total volatile fatty acids, propionate, and isovalerate concentrations, but a lower acetate:propionate ratio. The CTMR group had greater relative abundances of some cellulolytic bacteria (*Rikenellaceae* RC9 gut group, *Christensenellaceae* R7, *Ruminococcaceae* NK4A214, *Ruminococcaceae* UCG, *Ruminococcus*, and *Erysipelotrichaceae* UCG) in the rumen, which may be beneficial for the early acquisition of specific adult-associated microorganisms. In summary, a WPCS-based lactation TMR, but not the WPCS-included starter, had the potential to be an alternative starter in preweaning calves without having significant adverse effects. These findings provide theoretical and practical implications for the rational application of TMR in the early life of dairy calves.

Key words: corn silage, total mixed ration, growth performance, calf

INTRODUCTION

Calves are the basis of the dairy cow replacement herd; therefore, calf rearing has a significant effect on the sustainability and profitability of a dairy farm (Heinrichs and Heinrichs, 2011). One of the key goals among preweaning calves is to stimulate an early and healthy development of the rumen; this has been found to have a long-lasting effect on future performance (Soberon et al., 2012). Volatile fatty acids are mainly produced by rumen fermentation of cereals and have been previously confirmed to stimulate the development of rumen papilla (Lin et al., 2019; Chai et al., 2021). However, some studies have found that a large number of fermentable carbohydrates, especially starch, caused

Received July 3, 2022.
Accepted December 14, 2022.
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lower rumen pH and nutrient digestibility in preweaning calves, which further resulted in adverse effects on growth performance, health status, or both (Khan et al., 2016; Gelsinger et al., 2020). According to previous studies (Khan et al., 2011; Beiranvand et al., 2014; Xiao et al., 2020), providing a small amount of forage (approximately 10%) to preweaning calves can help to maintain rumen pH, prevent hyperkeratinization and clumping of the ruminal papillae, and improve health status. A ground starter supplemented with 10% (DM basis) chopped alfalfa hay produced more benefits in morphometric parameters of the rumen wall and rumen pH than did a textured or pelleted starter (Pazoki et al., 2017). However, these effects also relied on the sources, levels, and quality of the forage and the physical form of the starter diet (Imani et al., 2017; Kanani et al., 2019a,b).

Compared with hay supplementation, whole-plant corn silage (**WPCS**) can provide moisture to a diet and also has better digestibility owing to the shorter particle size (Kehoe et al., 2019). Higher moisture content in a calf diet increases both meal frequency and meal duration, resulting in longer eating periods and improved health-related variables, such as fecal score and general appearance score (Kargar and Kanani, 2019a,b). Calf diets containing 10% (DM basis) chopped alfalfa hay can be replaced with the same amount of WPCS without harming feed intake, growth performance, or meal pattern (Kanani et al., 2019b; Kargar and Kanani, 2019c; Kargar et al., 2019). Even though feeding WPCS alone stunted the growth of rumen papillae and tended to impair intestinal morphology, a mixture of starter and corn silage resulted in similar total villi and crypt depths compared with calves fed only starter (Kehoe et al., 2019). Regardless of the physical form of the starter, the inclusion of 15% (DM basis) WPCS in starter diets has been found to maintain rumen pH and improve the performance of preweaning dairy calves (Mirzaei et al., 2016).

Reducing the starch level of calf diets is another way to improve the rumen environment. Compared with steam-flaked corn, the inclusion of shredded sugar beet pulp, which contains highly digestible fiber but has a low starch content, in the starter diet had no significant effects on growth performance but increased rumination and eating time and decreased nonnutritive oral behavior (Kargar et al., 2021). Owing to their easy accessibility, high digestibility, and low cost, TMR, which also have a relatively low starch content, have been suggested as alternative solid feed sources for calves (Overvest et al., 2016; Kehoe et al., 2019). Based on the TMR concept, previous studies combined WPCS or reconstituted alfalfa hay with a calf starter as a mixed ration and found a similar effect on feed intake and growth performance in preweaning dairy calves (Kargar and Kanani, 2019c; Kargar et al., 2019). However, few studies have investigated the use of a standard WPCS-based lactation TMR, which has a significantly different particle size and chemical composition, including CP, NDF, and starch, compared with a calf starter meal, in preweaning calves. In addition, TMR can be a balanced and palatable source of nutrients for calves, which may be beneficial by leading to a smoother transition to solid feed (Overvest et al., 2016; Welk et al., 2022). The early exposure of calves to TMR, rather than the separate feeding of forage and starter meal, may have a beneficial influence on feeding behavior later in life by reducing feed sorting (Miller-Cushon et al., 2013; Xiao et al., 2018b).

In addition to VFA, the rumen microbiota also has a vital role in rumen development, digestibility, and health (Dias et al., 2018; Bi et al., 2019; Lin et al., 2019). The inoculation and establishment of an anaerobic ruminal microbial ecosystem are critical processes for triggering rumen development (Deelen et al., 2016). However, rumen microbial community changes have rarely been reported in calves supplemented with WPCS or fed a standard WPCS-based lactation TMR. Therefore, we hypothesized that WPCS inclusion and TMR feeding would have no significant effect on the growth performance of calves, but would change the rumen fermentation and microbial composition. To this end, we measured the feed intake, growth parameters, blood metabolites, rumen fermentation parameters, and ruminal microbial community in preweaning dairy calves fed either a starter including WPCS or a WPCS-based lactation TMR.

MATERIALS AND METHODS

This study followed the recommendations of the Instructive Notions with Respect to Caring for Experimental Animals, Ministry of Science and Technology of China. The protocol was approved by the Ethical Committee of the College of Animal Science and Technology of China Agricultural University (Beijing, China).

Animals, Housing, and Diet

This trial used 45 female Holstein calves (BW 38.0 ± 2.0 kg) born in late summer at the Modern Farming Group (Harbin, Heilongjiang, China). After birth, all the calves received 4 L of colostrum within 1 h. After a serum total protein (**TP**) test, only calves with a value of >5.5 g/dL were included in the following experiment. Calves were blocked by date of birth and assigned to 1 of 3 treatments that began at 2 d of age: a group fed 100% calf starter (**CONS**; n = 15); a group fed a mix

of 85% calf starter and 15% WPCS (**CSCS**; DM basis; n = 15); and a group fed 100% WPCS-based TMR (**CTMR**; n = 15; Table 1). The DM content of WPCS was 31.3%, and the starch, NDF, lactate, and butyrate contents were 28.4%, 46.0%, 8.4%, and 0.0% of DM, respectively (Supplemental Table S1, https://doi.org/10.6084/m9.figshare.20180984.v2; Zhang, 2022). The calves were housed in individual plastic hutches bedded with sand in the same barn from 2 d of age. Pasteurized normal milk (average milk protein and fat contents 3.37% and 3.98%, respectively; average total bacteria count and SCC <6,000 cfu/mL and <150,000 cells/mL, respectively) was fed twice a day at 0900 h and 1700 h from d 1 to d 14 at 4 L/meal, from d 15 to 29 at 5 L/meal, from d 30 to 49 at 6 L/meal, and from d 50 to 56 at 2 L/meal. From d 57 to 63, the amount of milk was reduced to 2 L/d and fed once a day, and then calves were weaned. After weaning, all the calves remained in the same hutches until d 70. Water was offered daily ad libitum.

Sample Collection and Analysis

Feed intake was measured daily during the experimental period. Samples of the calf starter, WPCS, and TMR were collected weekly, subsampled, and analyzed in the on-farm laboratory. The particle size fractions of the 3 treatment diets were evaluated by a Penn State Particle Separator (Kononoff et al., 2003). The physical effectiveness factor (**pef**) was calculated as the proportion of particles retained on the top 2 or 3 sieves ($pef_{>8}$ or $pef_{>4}$) of the separator. The physically effective NDF (**peNDF**) of the top 2 or 3 sieves ($peNDF_{>8}$ or $peNDF_{>4}$) was calculated by multiplying the NDF concentration of the feed by the fraction on $pef_{>8}$ or $pef_{>4}$, respectively. The in vitro nutrient digestibility of diets was measured according to the procedure described by Plaizier and Li (2013) with an Ankom Daisy II incubator (AD II; Ankom Technology Corporation). Briefly, the rumen fluid was collected 2 h after morning feeding from 3 ruminally fistulated lactating dairy cows fed the CTMR, squeezed through 4 layers of cheesecloth into a 39°C pre-warmed container, and immediately transferred to the laboratory. The culture buffer solution was prepared as described in a previous study (Menke and Steingass, 1988). Half a gram of samples (4 replicates per sample) was sealed in separate Ankom F57 bags (Ankom Technology Corporation) and placed in the fermentation bottle, which contained 2,400 mL of rumen fluid and buffer in a 1:2 (vol/vol) ratio. After a 24-h incubation, the bags were removed from the incubator, rinsed in cold running tap water until the wash water ran clear, and dried for nutrient measurement.

Table 1. Ingredients and nutrient compositions of the 3 treatment diets[1]

Item	Treatment		
	CONS	CSCS	CTMR
Ingredient, % of DM			
Alfalfa hay	—	—	1.8
Oat hay	—	—	3.5
Whole-plant corn silage	—	15.0	30.2
Soybean meal	17.4	14.8	6.3
Extruded soybean	19.9	16.9	1.3
Cottonseed meal	—	—	3.4
Cottonseed	—	—	6.4
DDGS	—	—	7.5
Corn gluten meal	5.9	5.0	—
Steam-flake corn	—	—	8.4
Ground corn	48.8	41.5	12.8
Corn germ meal	—	—	2.1
Corn gluten feed	—	—	0.7
Beet pulp	—	—	1.9
Molasses	2.4	2.0	2.6
Brewers grains	—	—	6.7
Wheat bran	—	—	0.2
Soybean hull	—	—	0.2
Fat powder	—	—	1.1
Calf premix[2]	5.7	4.8	—
Lactation premix[3]	—	—	3.0
Nutrient composition, % of DM			
DM	87.1	59.5	52.1
CP	20.3	18.8	16.6
Starch	34.1	34.5	26.0
NDF	11.0	15.4	30.4
ADF	5.2	8.2	17.0
EE	4.5	4.3	5.5
ME,[4] Mcal/kg of DM	3.6	3.3	3.0

[1]DDGS = distillers dried grains with soluble; EE = ether extract. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR.

[2]Purchased from Borui Feed; contained (per kg of premix; DM basis) 28.61% Ca, 325 mg of Cu, 5,500 mg of Zn, 4,000 mg of Mn, 4,000 mg of Se, 11.72 mg of I, 2.68 mg of Co, 400 mg of vitamin A, 150 mg of vitamin D_3, and 2,000 mg of vitamin E.

[3]Purchased from Borui Feed; contained (per kg of premix; DM basis) 0.6% Ca, 0.4% P, 38 mg of Cu, 123 mg of Zn, 89 mg of Mn, 1.24 mg of Se, 2.1 mg of I, 1.0 mg of Co, 17,000 IU of vitamin A, 6,700 IU of vitamin D_3, and 50,000 IU of vitamin E.

[4]Calculated according to NRC (2001) equations: ME = total digestible nutrients × 0.04409 × 0.82.

The contents of DM (method 930.15), CP (method 976.05), and ether extract (**EE**; method 920.39) in the diets were determined according to the procedures of AOAC International (2000). The NDF and ADF contents were determined according to a previous study, using a heat-stable α-amylase and sodium sulfite, and expressed including residual ash (Van Soest et al., 1991). Starch content was analyzed using a total starch assay kit (Megazyme) based on AOAC method 996.11 (AOAC International, 2000).

Body weight, withers height, body length, heart girth, and body barrel of each calf were measured on 2 consecutive days at the beginning and at wk 3, 5, 7,

and 9 of the experiment before the morning feeding. The mean values of the growth parameters were used to account for day-to-day variation.

Blood samples were collected from the jugular vein before the morning feeding on wk 3, 5, 7, and 9 from 5 randomly selected calves in each group. After sitting at room temperature for about 30 min, all blood samples were centrifuged at 3,000 × *g* at 4°C for 15 min to obtain plasma, which was separated into 1.5-mL tubes and stored at −20°C until further analysis. Blood metabolites, including glucose, TP, albumin, globulin, IgG, total cholesterol, total triglycerides, and urea nitrogen, were determined using commercial kits (Nanjing Jiancheng Bioengineering Institute) following the manufacturer's instructions.

Rumen fluid was sampled on d 40 and d 60 of age through a flexible esophageal tube (2-mm wall thickness and 6-mm internal diameter; Anscitech Co. Ltd.) from 15 calves (5 calves were randomly selected per treatment) at 2 h after the morning feeding. The pH of rumen fluid was measured with a mobile pH meter (Starter 300; Ohaus) immediately after collection. A rumen liquid fraction was obtained by filtering rumen fluid through 4 layers of cheesecloth followed by centrifugation (12,000 × *g* for 10 min at 4°C) to obtain a clear supernatant. One milliliter of filtered rumen fluid was analyzed for NH_3-N using a phenol-hypochlorite assay (Broderick and Clayton, 1997). Another aliquot of rumen fluid (1 mL) was combined with 100 μL of 25% metaphosphoric acid and then used to determine VFA according to a previous method (Zhang et al., 2020).

Microbial DNA Extraction, PCR Amplification, Sequencing, and Analysis

Genomic DNA was extracted from these 15 rumen samples (n = 5) on d 60 using an EZNA Stool DNA Kit (Omega Bio-Tek), following the manufacturer's protocol. After testing using 1% agarose gel and quantification using a Qubit 2.0 Fluorometer (Thermo Scientific), 30 ng of DNA was used for 25-μL PCR reaction mixtures. The V3 to V4 region of the bacteria 16S rRNA gene was amplified using forward (5′-GTACTCCTACGGGAGGCAGCA-3′) and reverse primers (5′-GTGGACTACHVGGGTWTCTAAT-3′; Song et al., 2017). The PCR conditions were as follows: an initial pre-denaturation at 94°C for 5 min, denaturation by 30 cycles of 94°C for 30 s, annealing at 55°C for 30 s, elongation at 72°C for 60 s, and then a final extension at 72°C for 7 min and holding at 4°C. Sequencing was performed on an Illumina MiSeq platform (2 × 250 bp) by Beijing Allwegene Technology Co. Ltd. (Beijing, China). After trimming the adaptor and primer sequences from Illumina reads, the raw sequences were assembled for each sample according to the unique barcode using QIIME (v.1.8, http://qiime.org/). Quality filtering was performed under specific filtering conditions to obtain high-quality clean tags according to QIIME. High-quality sequences were clustered into operational taxonomic units (**OTU**), defined as comprising sequences with <3.0% difference using UPARSE (v. 7.0, http://drive5.com/uparse/). After removing chimeric OTU with UCHIME (Edgar, 2013), the most abundant sequences within each OTU were selected as representative sequences and used for taxonomic classification and denomination based on the SILVA bacteria database (v. 123; http://www.arb-silva.de; Pruesse et al., 2007) using the RDP classifier with a 0.80 confidence threshold (Wang et al., 2007). Chao1, observed species, phylogenetic diversity whole tree, and Shannon were used to estimate α diversity. After assembling and filtering, the raw sequence was submitted to the NCBI Sequence Read Archive (http://www.ncbi.nlm.nih.gov/Traces/sra/), under accession number SUB11408050.

Statistical Analysis

The sample sizes of intake, growth performance, blood metabolites, rumen fermentation parameters, and microbial community were estimated to obtain a power of 0.8 under a significance level of 0.05. The feed intake, growth performance, blood metabolites, and rumen fermentation parameters, which were seen as repeated measurements, were first checked for normality and analyzed using the mixed procedure of SAS 9.4 (SAS Institute Inc.) with week as a repeated effect. The model included treatment, week, and the interaction between them as fixed effects, and calf within treatment as a random effect. Covariance structures, including autoregressive type 1, compound symmetry, and unstructured, which had the lowest Akaike information criterion, were used in the covariance structures model (Littell et al., 1998). The initial BW, initial body structural measurements, and feed intake were included as covariates in the statistical analysis for average BW, average body structural, and ADG, respectively. However, the initial BW and body structural measurements did not improve the significance ($P > 0.05$) of the model and were subsequently removed. Results were reported as least squares means. Variables of microbial α diversity were analyzed with the Kruskal–Wallis test. Analysis of similarities was used to test for significant differences between treatments. The level of statistical significance was set at $P < 0.05$. A tendency for significance was declared at $0.05 \leq P < 0.10$. Variables of microbial communities were analyzed with linear discriminant analysis effect size analysis to identify

Table 2. Particle size fractions and in vitro digestibility of the 3 treatment diets[1]

Item	Treatment			SEM	*P*-value
	CONS	CSCS	CTMR		
Particle size fraction, % as-fed basis[2]					
Long (>19 mm)	0.00[c]	2.06[b]	7.05[a]	1.057	<0.01
Medium (<19, >8 mm)	43.37[b]	74.11[a]	40.67[b]	5.409	<0.01
Short (<8, >4 mm)	56.63[a]	19.77[b]	15.05[c]	6.594	<0.01
Fine (<4 mm)	0.00[c]	4.07[b]	37.23[a]	5.899	<0.01
$pef_{>8}$	43.37[b]	76.17[a]	47.72[b]	5.180	<0.01
$pef_{>4}$	100.00[a]	95.93[b]	62.77[c]	5.899	<0.01
$peNDF_{>8}$	4.76[c]	11.74[b]	14.51[a]	1.459	<0.01
$peNDF_{>4}$	11.00[c]	14.78[b]	19.08[a]	1.178	<0.01
In vitro digestibility, %					
DM	78.37[a]	67.39[b]	61.10[c]	0.914	<0.01
CP	72.87[a]	52.91[b]	47.11[c]	1.407	<0.01
EE	65.90[a]	54.22[b]	54.86[b]	0.842	<0.01
NDF	59.89[a]	45.00[b]	27.08[c]	2.457	<0.01
ADF	54.63[a]	38.42[b]	22.53[c]	1.662	<0.01
Starch	89.70	89.71	89.67	0.013	0.32

[a–c]Means within a row with different superscripts differ significantly ($P < 0.05$).

[1]EE = ether extract; pef = physical effectiveness factor; peNDF = physically effective NDF. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR.

[2]Particle size fraction (as-fed basis) was determined by a Penn State Particle Separator (Kononoff et al., 2003). Subscripts >8 and >4 refer to proportions of contents retained on the top 2 or 3 sieves.

significantly changed bacteria between groups using the criterion of linear discriminant analysis score >3.0 (Segata et al., 2011).

RESULTS AND DISCUSSION

Diet Particle Size Fraction and In Vitro Digestibility

The CONS diet contained neither long (>19 mm) nor fine (<4 mm) particles (Table 2). The CTMR group had more ($P < 0.01$) long and fine particles than the CSCS group. Medium (<19, >8 mm) particles were more abundant ($P < 0.01$) in CSCS than in CONS and CTMR, whereas short (<8, >4 mm) particles were least ($P < 0.01$) in CTMR and greatest in CONS. The particle size distributions of diets were mostly related to the physical characteristics of ingredients (Kargar and Kanani, 2019c; Kargar et al., 2021). To diminish the effects of dustiness on the respiratory tract of calves, it is important to minimize the dust content in starter meals. Compared with CONS, the particle size fraction of CSCS was significantly affected by WPCS inclusion. In this study, the WPCS was harvested at a theoretical length of cut of 19 mm, which provided both fiber effectiveness and digestibility (Salvati et al., 2017). The particle size distribution of CTMR was similar to those used in previous studies on lactating dairy cows, with more fractions of medium and fine particles in diets with WPCS (Kmicikewycz et al., 2015; Castro et al., 2019). Despite CTMR having lower ($P < 0.01$) values of $pef_{>8}$ and $pef_{>4}$, the values of $peNDF_{>8}$ and $peNDF_{>4}$ were greatest in CTMR and lowest in CONS owing to the NDF content.

The 24-h in vitro digestibility of DM, CP, NDF, and ADF were the lowest in CTMR and greatest in CONS with CSCS in the middle ($P < 0.01$); this was likely owing to more long particles and fiber-rich ingredients as well as byproduct meals in CTMR. The digestibility of EE was lower ($P < 0.01$) in CSCS and CTMR than in CONS. Despite the starch level being lower in CTMR, the 3 treatments had similar digestibility of starch, which might be attributed to the similar starch sources from corn.

Feed Intake and Growth Parameters

As experimental calves consumed almost all the milk offered, the nutrient intake from milk was similar among treatments (Table 3). Because CTMR had greater NDF and ADF content in the diet, it seems reasonable to expect a lower DMI than in CONS and CSCS. However, the similar DMI among treatments, or even numerically greater DMI in CTMR, led us to reconsider the effects of physical type and chemical composition of TMR on DMI. A TMR is an even mixture of forage, concentrate, and water; in this study it comprised WPSC, steam-flaked corn, and molasses; such a diet has been shown to be palatable for calves (Welk et al., 2022). Similarly, increasing the moisture content of solid feed by using reconstituted alfalfa hay can increase DMI in preweaning calves (Kargar and Kanani, 2019b). Compared with a previous study that fed silage–based TMR

Table 3. Nutrient intake from milk and solid feed in preweaning dairy calves (n = 15)[1]

Item	Treatment			SEM	*P*-value		
	CONS	CSCS	CTMR		Treat	Week	Treat × week
Nutrient intake from milk							
DM, kg/d	1.1	1.1	1.1	0.02	0.87	<0.01	0.99
CP, g/d	299.7	298.9	299.2	1.04	0.93	0.08	0.99
Fat, g/d	354.0	353.0	353.4	1.71	0.85	<0.01	0.96
ME,[2] Mcal/d	5.6	5.6	5.6	0.11	0.87	<0.01	0.99
Nutrient intake from solid feed							
DM, g/d	245.5	246.9	293.7	20.19	0.16	<0.01	0.90
CP, g/d	49.8	46.4	48.8	3.60	0.79	<0.01	0.96
Starch, g/d	83.6	85.2	76.4	6.05	0.55	<0.01	0.99
NDF, g/d	27.0[b]	37.9[b]	89.3[a]	4.98	<0.01	<0.01	<0.01
ADF, g/d	12.6[b]	20.2[b]	49.9[a]	2.76	<0.01	<0.01	<0.01
EE, g/d	11.0[b]	10.6[b]	16.2[a]	1.02	<0.01	<0.01	0.04
ME,[2] Mcal/d	0.9	0.8	0.9	0.06	0.70	<0.01	0.95
Total nutrient intake[3]							
DM, kg/d	1.3	1.3	1.4	0.02	0.19	<0.01	0.90
CP, g/d	350.7	347.2	349.6	4.80	0.79	<0.01	0.96
EE, g/d	365.2[b]	364.9[b]	370.4[a]	3.05	<0.01	<0.01	0.04
ME,[2] Mcal/d	6.5	6.4	6.5	0.09	0.67	<0.01	0.96
Potentially digestible nutrient intake from solid feed							
DM, g/d	192.37	166.40	179.42	12.414	0.38	<0.01	0.79
CP, g/d	36.31[a]	24.55[b]	22.96[b]	1.957	<0.01	<0.01	<0.01
Starch, g/d	74.97	76.42	68.46	5.054	0.54	<0.01	0.99
NDF, g/d	16.17[b]	17.06[b]	24.17[a]	1.388	<0.01	<0.01	0.10
ADF, g/d	6.91[b]	7.74[b]	11.25[a]	0.634	<0.01	<0.01	0.03
EE, g/d	7.26[ab]	5.77[b]	8.86[a]	0.524	<0.01	<0.01	0.05

[a,b]Means within a row with different superscripts differ significantly ($P < 0.05$).

[1]EE = ether extract. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR. Treat = treatment.

[2]Calculated according to NRC (2001) equations: ME = total digestible nutrients × 0.04409 × 0.82.

[3]Calculated as the sum from milk and solid feed.

to preweaning calves (Overvest et al., 2016), the lower ADF content and milk allowance might have contributed to the greater DMI from the solid feed in the CTMR group in this study. Owing to the greater NDF, ADF, and EE contents in the diet and similar solid feed intakes, CTMR had greater ($P < 0.01$) NDF, ADF, and EE intakes from solid feed than did CONS and CSCS. Because the DM and CP intakes were mainly from milk and the 3 treatments had similar milk intakes, the total DM and CP intakes were similar. The total EE intake was still greater ($P < 0.01$) in CTMR than in CONS and CSCS. Because milk contains no starch, NDF, or ADF, the intakes of these 3 nutrients from solid feed were also the total. Numerically, calves received more ME from liquid milk than from solid feed. The total ME intake was similar among treatments, which might have contributed to the similar growth performance, consistent with previous studies including WPCS and feeding TMR in preweaning calves (Overvest et al., 2016; Iqbal et al., 2019).

The potentially digestible nutrient intake from solid feed was related to both intake and digestibility. The greater ($P < 0.01$) potentially digestible NDF, ADF, and EE intakes from the solid feed in the CTMR group indicated that intake had more influence than digestibility on the potentially digestible nutrient intake. Compared with CONS, the lower ($P < 0.01$) potentially digestible CP intake was attributed to the similar CP intake but lower CP digestibility in CONS and CSCS.

Consistent with the previous standard for weaning (Hulbert and Moisa, 2016), milk supplementation started to be reduced when calves were at about 50 d old at a body weight of twice their birth weight. Compared with CONS, CSCS calves had a larger ($P = 0.03$) body barrel, followed by CTMR (Table 4). The different feed types did not affect ($P > 0.05$) body weight, withers height, body length, hip height, heart girth, cannon bone circumference, ADG, or feed efficiency (ADG/total DMI and ADG/total ME intake). Similarly, replacing steam-flaked corn with shredded sugar beet pulp and having a low-starch solid feed did not have an adverse effect on calf performance (Kargar et al., 2021). It is worth noting that the greater body barrel in CSCS might be related to greater gut fill resulting from greater NDF and ADF intakes but lower NDF and ADF digestibility compared with CONS. However, we could not measure the gut fill in an on-farm situation. Moreover, CTMR calves were expected

Table 4. Effects of different feeding on overall growth parameters in dairy calves (n = 15)[1]

Item	Treatment			SEM	*P*-value		
	CONS	CSCS	CTMR		Treat	Week	Treat × week
BW, kg	62.83	62.63	62.36	1.022	0.85	<0.01	1.00
Withers height, cm	83.90	82.82	83.63	0.396	0.12	<0.01	0.42
Body length, cm	80.06	79.56	79.11	0.447	0.78	<0.01	0.95
Hip height, cm	88.13	87.84	88.53	0.389	0.45	<0.01	0.44
Heart girth, cm	93.93	94.40	93.72	0.404	0.46	<0.01	0.66
Body barrel, cm	100.07[b]	101.86[a]	100.42[ab]	0.514	0.03	<0.01	0.81
Cannon bone circumference, cm	11.77	11.87	11.83	0.083	0.66	<0.01	0.20
ADG, kg/d	0.78	0.77	0.78	0.032	0.95	<0.01	1.00
ADG/TDMI	0.60	0.63	0.62	0.021	0.82	<0.01	0.97
ADG/TMEI, kg/Mcal	0.12	0.13	0.13	0.004	0.48	<0.01	0.82

[a,b]Means within a row with different superscripts differ significantly ($P < 0.05$).

[1]TDMI = total DMI; TMEI = total ME intake. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR. Treat = treatment.

to have the greatest body barrel considering the dietary characteristics of CTMR, especially in NDF and ADF intakes and digestibility. The different result might be attributed to a greater rumen passage rate, as high-fiber and low-starch diets accelerate the passage rate and result in reduced retention times (Pino et al., 2018; Erickson et al., 2020). Future studies regarding the effects of dietary NDF and ADF intakes and digestibility on gut fill and rumen passage rate in preweaning calves are warranted.

Blood Metabolites and Rumen Fermentation Parameters

As important biomarkers, TP, albumin, and globulin in the blood can reflect both health status and protein utilization in animals (Liu et al., 2021). Although the albumin concentration tended to be greater ($P = 0.05$) in CSCS than in CONS (Table 5), the values of TP, albumin, and globulin concentrations, as well as the albumin:globulin ratio, were in the normal range of previously reported preweaning calves (Zou et al., 2017; Liu et al., 2021), which indicated that all the experimental animals were healthy and had similar CP intakes. All the treatments had IgG concentrations of greater than 10 g/L, indicating successful passive transfer of maternal IgG (Deelen et al., 2014). In agreement with our study, the forage type, feeding method, and physical form of starter-based TMR did not significantly affect most blood metabolites in preweaning calves (Kanani et al., 2019b; Engelking et al., 2020).

The CSCS group had lower ($P \leq 0.02$) rumen pH and total VFA concentration than CTMR, followed by CONS (Table 6). Rumen pH in CSCS was lower than 5.6, which indicated a higher risk of subacute rumen acidosis (Gelsinger et al., 2020). However, rumen pH commonly falls below 5.6 in calves fed diets containing grains or forages, especially during weaning (Laarman et al., 2012; Suarez-Mena et al., 2015, 2016). The higher rumen pH in CTMR might have been brought about by greater salivary secretion, which reflects more time spent on rumination, caused by the lower starch

Table 5. Effects of different feeding on blood metabolites in dairy calves (n = 5)[1]

Item	Treatment			SEM	*P*-value		
	CONS	CSCS	CTMR		Treat	Week	Treat × week
Glucose, mmol/L	4.57	4.68	4.79	0.126	0.49	0.12	0.18
Total protein, g/L	45.28	47.11	47.89	1.133	0.26	<0.01	0.10
Albumin,[2] g/L	21.32	23.10	22.76	0.530	0.05	<0.01	0.07
Globulin, g/L	23.96	24.01	25.14	0.860	0.55	0.52	0.31
Albumin:globulin	0.91	0.98	0.91	0.032	0.23	<0.01	0.74
IgG, g/L	10.22	10.24	10.57	0.195	0.37	0.39	0.03
TC, mmol/L	2.04	2.09	2.25	0.070	0.31	<0.01	0.83
TG, mmol/L	0.69	0.61	0.70	0.046	0.27	<0.01	0.53
Urea, mmol/L	2.60	2.76	2.42	0.126	0.17	<0.01	0.02

[1]TC = total cholesterol; TG = total triglyceride. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR. Treat = treatment.

[2]There was a different trend between the CONS and CSCS groups ($P = 0.06$).

Table 6. Effects of different feeding on rumen fermentation parameters in dairy calves (n = 5)[1]

	Treatment				*P*-value		
Item	CONS	CSCS	CTMR	SEM	Treat	Week	Treat × week
pH	5.59[ab]	5.36[b]	5.94[a]	0.11	<0.01	0.76	0.60
NH_3-N, mg/dL	10.56	7.24	8.34	1.47	0.29	0.33	0.96
TVFA, mmol/L	95.56[ab]	77.20[b]	108.64[a]	7.42	0.02	<0.01	0.08
Acetate, mmol/L	42.81	39.65	49.94	3.28	0.08	<0.01	0.33
Propionate, mmol/L	38.55[a]	25.96[b]	47.30[a]	3.56	<0.01	<0.01	0.01
Butyrate, mmol/L	9.22	8.05	6.34	2.02	0.57	0.49	0.51
Valerate, mmol/L	4.71[a]	2.83[b]	3.88[ab]	0.44	0.01	0.01	0.52
Isovalerate, mmol/L	0.96[ab]	0.42[b]	1.37[a]	0.17	<0.01	<0.01	<0.01
A:P	1.16[b]	1.63[a]	1.21[b]	0.08	<0.01	<0.01	0.07

[a,b]Means within a row with different superscripts differ significantly ($P < 0.05$).

[1]A:P = acetate-to-propionate ratio; TVFA = total VFA. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR. Treat = treatment.

content and greater $peNDF_{>4}$ value compared with the other 2 groups. Previous studies also report increased rumen pH in cows with low starch and high peNDF levels, which induced longer ruminating and total chewing times (Beauchemin and Yang, 2005; Cao et al., 2021).

Rumen VFA, especially propionate and butyrate, presumably contribute to the growth of the ruminal epithelium (Baldwin et al., 2004; Yohe et al., 2019). Compared with CONS and CTMR, CSCS had a greater ($P < 0.01$) acetate:propionate ratio but lower ($P < 0.01$) propionate concentration, indicating that CSCS might not be superior in stimulating rumen development compared with the other 2 groups. The CSCS had a lower ($P = 0.01$) valerate concentration than CONS and had a lower ($P < 0.01$) isovalerate concentration than CTMR. Different feed types had no significant effect on butyrate concentration. Compared with only feeding starter, the inclusion of corn silage impaired rumen papillae length and width, while increasing papillae concentration (Kehoe et al., 2019). The greater rumen pH, total VFA concentration, and propionate concentration might suggest a better rumen environment and development in CTMR. The greater propionate concentration in CTMR might be attributable to greater fine particle fraction as well as relatively higher starch digestibility than NDF and ADF digestibility (89.67% vs. 27.08% and 22.53%) in the rumen. However, more studies investigating rumen morphology need to be performed to correctly describe rumen development.

Rumen Bacterial Diversity and Community

After rarefaction analysis, 25,440 sequences per sample were used for diversity analysis. After OTU picking and chimera checking, 807 OTU were calculated for all the samples at 3% dissimilarity. The numbers of OTU for CONS, CSCS, and CTMR were 291, 273, and 762, respectively; furthermore, 192 OTU were found in all 3 treatments (Supplemental Figure S1, https://doi.org/10.6084/m9.figshare.20180984.v2; Zhang, 2022). Good's coverage for each sample was >0.99. Compared with CONS and CSCS, CTMR had a greater Chao1 value, observed species, and phylogenetic diversity whole tree (Figure 1). We found that CTMR had the greatest Shannon index, followed by CSCS, and CONS had the lowest Shannon index. Nonmetric multidimensional scaling of each group revealed that the samples were clustered by treatments (Figure 2). Analysis of similarities showed significant differences in bacterial community composition between treatments CONS and CSCS (r = 0.55, $P < 0.01$) and between treatments CONS and CTMR (r = 0.54, $P = 0.03$). A tendency of difference was found between treatments CSCS and CTMR (r = 0.37, $P = 0.06$).

A total of 16 phyla were detected via taxonomic analyses. The 5 most abundant phyla were *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, *Proteobacteria*, and *Saccharibacteria*, representing 49.41, 34.39, 8.58, 3.39, and 2.50% of total sequences, respectively (Figure 3A), which was similar to findings from previous studies (Dill-McFarland et al., 2017; Xiao et al., 2018a; Malmuthuge et al., 2019). At the genus level, 168 bacterial genera were detected, and 26 genera were common to all samples; these were identified as the core bacterial microbiome (Supplemental Figure S2, https://doi.org/10.6084/m9.figshare.20180984.v2; Zhang, 2022). Consistent with previous studies (Dias et al., 2018; Cersosimo et al., 2019; Li et al., 2019), *Prevotella* 7 (10.40% of 16S rRNA gene reads), *Lachnospiraceae* NK3A20 (9.95%), *Olsenella* (7.97%), *Prevotella* 1 (7.93%), *Succiniclasticum* (7.51%), *Acetitomaculum* (3.10%), *Megasphaera* (3.04%), *Syntrophococcus* (2.68%), Candidatus *Saccharibacteria bacterium* UB2523 (2.35%), and *Mitsuokella* (2.27%) were the top 10 most abundant genera (Figure 3B).

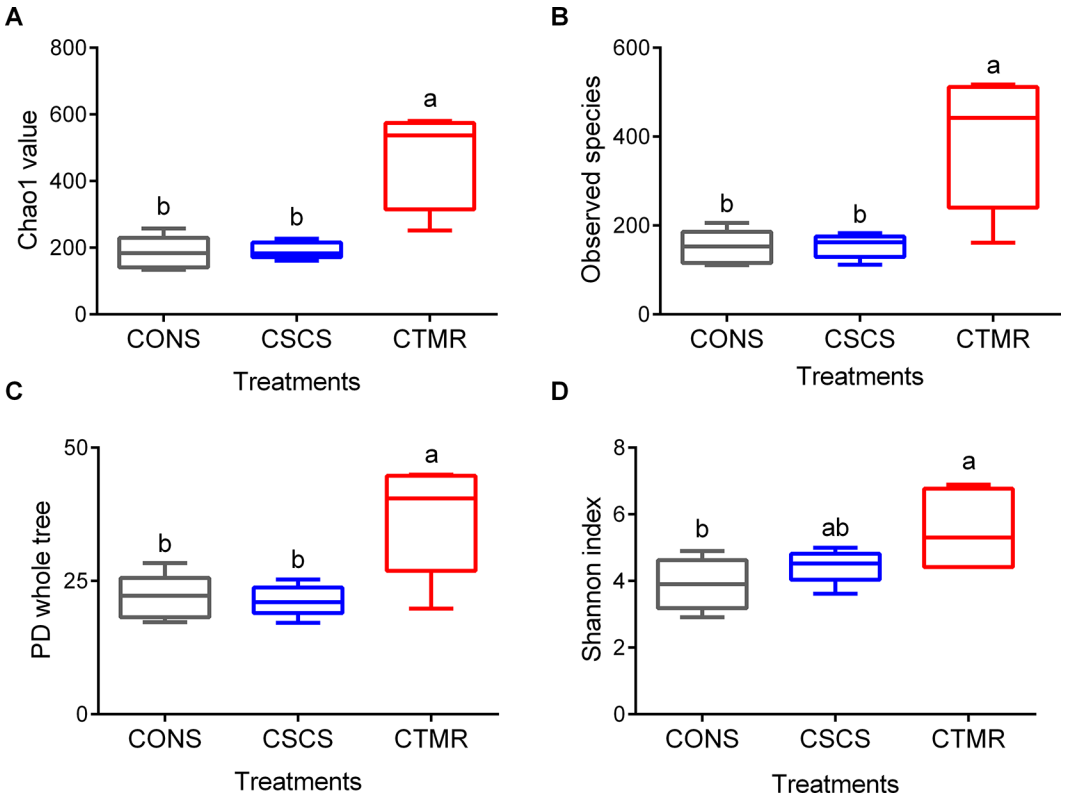


Figure 1. Effects of different feeds on the rumen bacterial α diversity in dairy calves: (A) bacterial Chao1 value, (B) observed species, (C) phylogenetic diversity (PD) whole tree, and (D) Shannon index. Boxes represent the interquartile range (IQR) between the first and third quartiles (25th and 75th percentiles, respectively), and the horizontal line inside the box defines the median. Whiskers represent the lowest and highest values within 1.5 times the IQR from the first and third quartiles, respectively. Boxes with different letters above their whiskers are significantly different ($P < 0.05$) among treatments. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR.

Moreover, these bacteria also represent the main bacterial taxa in the gastrointestinal tracts of dairy cattle (Henderson et al., 2015; Zhang et al., 2017; Zhang et al., 2018).

Linear discriminant analysis effect size analysis showed that 51 taxa were significantly changed among treatments (Supplemental Figure S3, https://doi.org/10.6084/m9.figshare.20180984.v2; Zhang, 2022).

Table 7. Relative abundances (%) of main significant bacterial genera among treatments (n = 5)[1]

Item	Taxa (phylum; family; genus)	Relative abundance in treatment, %			LDA score ($\log_{10}$)
		CONS	CSCS	CTMR	
CONS enriched	*Firmicutes*; *Lachnospiraceae*; *Acetitomaculum*	7.15	1.14	1.00	4.47
CONS enriched	*Firmicutes*; *Lachnospiraceae*; *Oribacterium*	0.78	0.34	0.18	3.47
CONS enriched	*Firmicutes*; *Lachnospiraceae*; *Syntrophococcus*	5.78	1.53	0.74	4.41
CONS enriched	*Firmicutes*; Family XIII; Family XIII UCG 001	0.11	0.29	0.04	3.10
CONS enriched	*Firmicutes*; *Lachnospiraceae*; *Roseburia*	0.32	0.81	0.20	3.47
CONS enriched	*Firmicutes*; *Acidaminococcaceae*; *Acidaminococcus*	0.38	0.60	0.08	3.36
CONS enriched	*Actinobacteria*; *Coriobacteriaceae*; *Denitrobacterium*	0.01	0.02	<0.01	3.28
CTMR enriched	*Bacteroidetes*; *Prevotellaceae*; *Prevotellaceae* UCG	0.19	0.03	2.59	3.64
CTMR enriched	*Bacteroidetes*; *Rikenellaceae*; *Rikenellaceae* RC9 gut group	0.07	0.04	2.07	3.91
CTMR enriched	*Firmicutes*; *Christensenellaceae*; *Christensenellaceae* R7	0.01	0.01	1.47	3.79
CTMR enriched	*Firmicutes*; *Ruminococcaceae*; *Ruminococcaceae* NK4A214	0.15	0.07	2.05	4.00
CTMR enriched	*Firmicutes*; *Ruminococcaceae*; *Ruminococcaceae* UCG	0.13	0.03	1.43	3.36
CTMR enriched	*Firmicutes*; *Ruminococcaceae*; *Ruminococcus*	0.03	0.03	1.27	3.78
CTMR enriched	*Firmicutes*; *Erysipelotrichaceae*; *Erysipelotrichaceae* UCG	<0.01	<0.01	1.80	3.99

[1]LDA = linear discriminant analysis. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR.

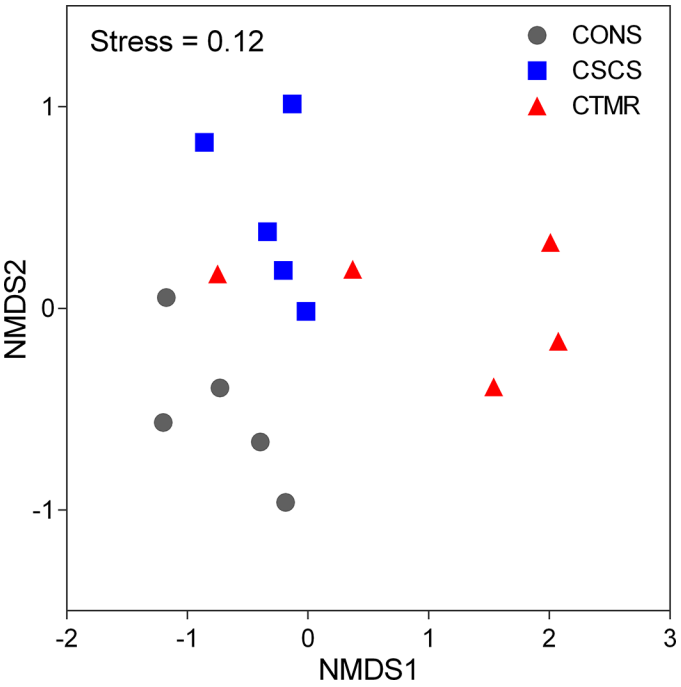


Figure 2. Similarity of the bacterial communities among treatments. Distances between the samples were calculated using the Bray-Curtis similarity index, based on similarity in operational taxonomic units (OTU) composition (OTU similarity ≥0.97%) of each sample, and plotted using nonmetric multidimensional scaling (NMDS). Each point represents a different sample plotted according to its OTU composition and abundance (stress value = 0.12). A greater distance between 2 points implies a lower similarity, whereas samples with a more similar bacterial composition and abundance cluster closer together. CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR.

Among these, 14 genera present in more than half of the samples were observed to be the main significant genera among treatments (Table 7). Even though different feed types had a limited effect on rumen bacteria at the phylum level, bacteria at the genus level were altered. One of the most abundant bacteria at the genus level, *Acetitomaculum* was highly enriched in CONS. According to previous studies (Greening and Leedle, 1989; Rees et al., 1995; Liu et al., 2019), *Acetitomaculum* can use hydrogen to produce acetate in the rumen. Similarly, as cellulolytic bacteria, *Syntrophococcus* can synthesize acetate from one-carbon compounds (Doré and Bryant, 1990). However, the similar ruminal acetate concentrations were inconsistent with the relative abundances of *Acetitomaculum* and *Syntrophococcus*, which indicated that other acetogenins such as *Clostridium*, *Blautia*, *Lachnospiraceae*, and *Ruminococcaceae* might have been involved in ruminal acetate production in this study (Yang et al., 2016).

With limited studies on *Rikenellaceae*, the metabolic function of the *Rikenellaceae* RC9 gut group in the rumen has not yet been fully defined (Ramos et al., 2018); however, it is known that their metabolic end products include acetate, propionate, succinate, and alcohol (Prokofeva et al., 2014). Nevertheless, several studies report that dietary starch decreases the relative abundance of the *Rikenellaceae* RC9 gut group, indicating that this family may be involved in the primary or secondary degradation of structural carbohydrates (Pitta et al., 2010; Zened et al., 2013; Ramos et al., 2018); our study had similar findings, in that CTMR with a lower starch diet content had a greater relative abundance of the

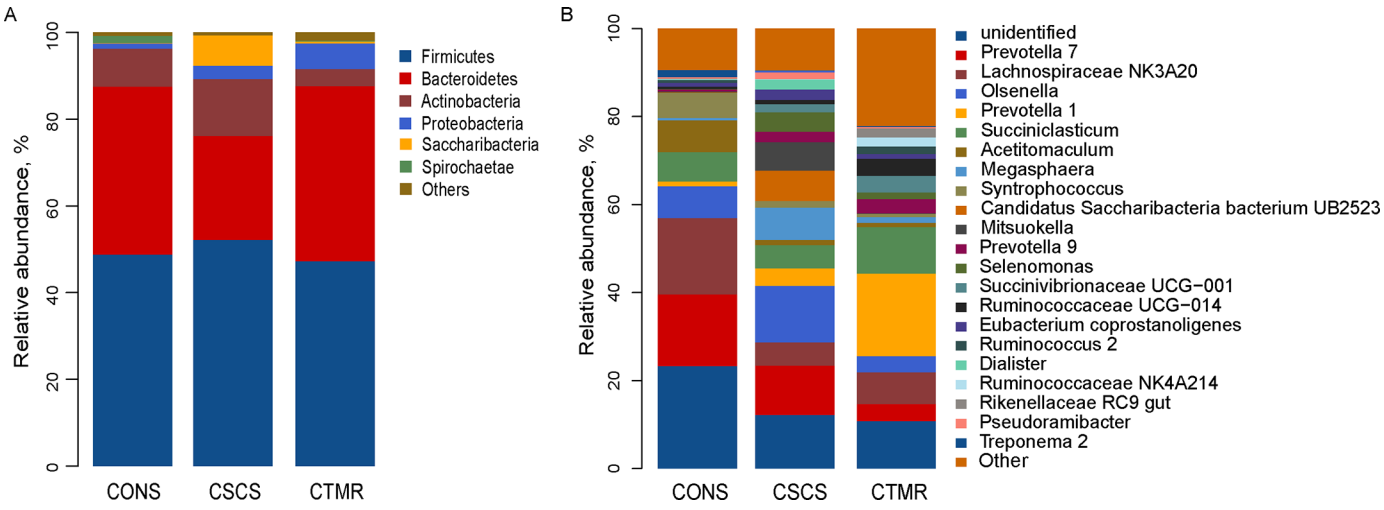


Figure 3. Effects of different feeds on changes of ruminal microbial taxa at the phylum (A) and genus levels (B). CONS = 100% calf starter; CSCS = a mix of 85% calf starter and 15% whole-plant corn silage (DM basis); CTMR = 100% whole-plant corn silage–based TMR.

Rikenellaceae RC9 gut group. *Christensenellaceae* are ubiquitous bacteria in the gastrointestinal tract of humans and animals (Waters and Ley, 2019). It contains α-arabinosidase, β-galactosidase, and β-glucosidase and is predominant in diets including chicory and chitosan (Li et al., 2016, 2020; Perea et al., 2017); we can therefore assume that members of this family can utilize nonfibrous carbohydrate. Previous studies also suggest that *Rikenellaceae* and *Christensenellaceae* might be related to lipid metabolism and associated with reduced visceral adipose tissue and healthier metabolic profiles (Morotomi et al., 2012; Waters and Ley, 2019).

A genome comparison study indicated that *Ruminococcaceae* have glycoside hydrolases and carbohydrate-binding modules, significantly enriched in endo-1, 4-β-xylanase, and cellulase genes (Biddle et al., 2013). As a representative of *Ruminococcaceae*, *Ruminococcus flavefaciens* has unusual polysaccharide binding and degradation strategies; *Ruminococcaceae* members are also well known to be present in the rumen and to possess cellulolytic activity (Russell, 2009; Paz et al., 2018; Sun et al., 2019). Furthermore, *Ruminococcaceae* and unclassified *Bacteroidales* have been found to be more abundant in the rumen of animals fed high-forage diets (Henderson et al., 2015). Belonging to the same family of *Ruminococcaceae*, *Ruminococcaceae* NK4A214, *Ruminococcaceae* UCG, and *Ruminococcus* might also have similar capacities for fiber digestion (Qian et al., 2019; Wang et al., 2019), which aligns with the greater potentially digestible NDF and ADF intakes and the numerically lower potential digestible starch intake in the CTMR group. Interestingly, some studies have also found that *Ruminococcaceae* family members are crucial to butyrate production or are positively correlated with ruminal propionate content (Bui et al., 2016; Pan et al., 2017), which is consistent with the greater propionate content and relative abundance of *Ruminococcaceae* UCG in the CTMR group of the current study.

Although *Erysipelotrichaceae* have been reported in multiple ecological environments, their function has not been well illustrated. In humans and mice, *Erysipelotrichaceae* are likely related to lipid or cholesterol phenotypes and are increased in diets containing high levels of beef or fat (Kaakoush, 2015; Nagao-Kitamoto et al., 2016). In ruminants, the cooperation between *Erysipelotrichaceae* UCG and *Sphaerochaeta* may benefit the degradation of plant fibers in the rumen (Qian et al., 2019). It is also reported that the relative abundance of *Erysipelotrichaceae* increases in older dairy calves with a more diverse and stable microbial community (Song et al., 2018). The lipid and fiber degradation properties of *Erysipelotrichaceae* followed the nutrient composition in the diet of the CTMR group. Most of these enriched bacteria in the CTMR group are involved in cellulose utilization and are important in adult cattle as fiber-degrading organisms (Henderson et al., 2015; Mao et al., 2015; Xue et al., 2022); furthermore, their presence might suggest the early acquisition of specific adult-associated microorganisms in TMR-fed calves. Compared with the CSCS group, the greater relative abundance of cellulolytic bacteria in the CTMR group might also be related to these calves' higher rumen pH, as the growth of cellulolytic bacteria would be impaired by a reduction in rumen pH in the CSCS group (Russell and Wilson, 1996).

CONCLUSIONS

Despite the different characteristics of diets among treatments, CTMR had increased fiber intake and relative abundances of some ruminal cellulolytic bacteria, and improved rumen fermentation without adverse effects on DMI, blood metabolites, and growth parameters in preweaning calves compared with the other 2 groups; these findings may be beneficial for postweaning development. Therefore, we suggest that calf starter can be replaced with a lactation TMR in preweaning calves; however, a WPCS-included starter cannot be recommended, at least under these experimental conditions.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (grant numbers 32102570 and 32130100; Beijing, China), the Key R&D Program of Shaanxi Province (grant number 2022ZDLNY01-11; Xi'an, China), the Chinese Universities Scientific Fund (grant number 2452020188; Yangling, China), and the National Dairy Industry and Technology System of China (grant number CARS-36; Beijing, China). We thank the Modern Farming Group (Harbin, Heilongjiang, China) for allowing us to use their animals and facilities. The authors have not stated any conflicts of interest.

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